

ORF307

Constrained least squares

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Today's topics

- Linearly constrained least squares
- Solving the constrained least squares
- Portfolio optimization

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- **Linearly least squares**

Least squares with equality constraints

Definition: $\min ||Ax - b||^2$
s.t. $Cx = d$

Problem data: $A \in R^{m \times n}$ $b \in R^m$
 $C \in R^{p \times n}$, $d \in R^p$

$$Cx = d \Leftrightarrow c_i^T x = d_i, i = 1, 2, \dots, p$$

Definitions:

- **Feasible region:** $\mathcal{F} = \{x \in R^n : Cx = d\}$
- $x \in R^n$ is a **feasible solution** if $x \in \mathcal{F}$
- $x^* \in R^n$ is an **optimal solution** if
 - $x^* \in \mathcal{F}$
 - $||Ax^* - b||^2 \leq ||Ax - b||^2, \forall x \in \mathcal{F}$

Interpretations:

- Solve least squares **together** with linear equations
- **Bi-objective least squares** with $\lambda = \infty$

$$\min J_1(x) + \lambda J_2(x) = ||Ax - b||^2 + \lambda ||Cx - d||^2$$

Optimal advertising (*recall the problem*)

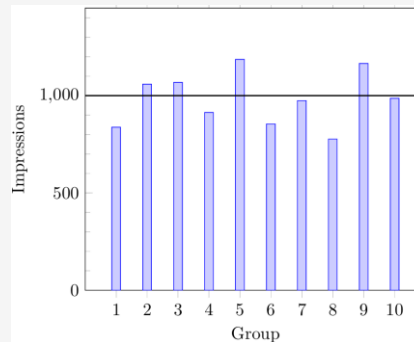
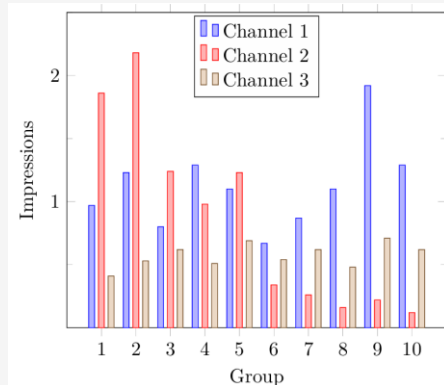
- m **demographic** groups (e.g., 18-22 yo, 23-33 yo, 34-44 yo, 45-65 yo, etc.)
- n **advertising** channels (e.g., ABC, CBS, CNN, Fox, NBC, PBS, BBC, DW, etc.)
- $A \in R^{m \times n}$ the **reach matrix** $\rightarrow$ contains the demographic reach of the channels
 - A_{ij} the number of **views** per \$ spent of demographic group i at the channel j
- $v^{des} \in R^m$ the vector of **desired views** for each demographic
- $s \in R^n$ is the **spending** on the n advertising channels (*not known*)
- $v = As \in R^m$ the **views** across the demographic channels (*not known*)
- **Goal:** Find s such that $\|As - v^{des}\|^2$ from the desired views is minimized

Optimal advertising *without budget constraint*

- Some representative data

- $m = 3, n = 10, v^{des} = (1000, 1000, 1000)^T$

$$A = \begin{bmatrix} 0.97 & 1.86 & 0.41 \\ 1.23 & 2.18 & 0.53 \\ 0.80 & 1.24 & 0.62 \\ 1.29 & 0.98 & 0.51 \\ 1.10 & 1.23 & 0.69 \\ 0.67 & 0.34 & 0.54 \\ 0.87 & 0.26 & 0.62 \\ 1.10 & 0.16 & 0.48 \\ 1.92 & 0.22 & 0.71 \\ 1.29 & 0.12 & 0.62 \end{bmatrix}$$



- Least squares

$$\min \|Ax - v^{des}\|^2$$

- Optimal spending: $x^* = (62, 100, 1443)^T$
- RMS Error: 13.20%
- Total cost: $\mathbf{1}^T x^* = \$1605$

Optimal advertising *with budget constraint*

- **Some representative data**

- $m = 3, n = 10, v^{der} = (1000, 1000, 1000)^T$

- **And total budget $B = \$1284$**

Scale unconstrained least squares solution to match budget B :

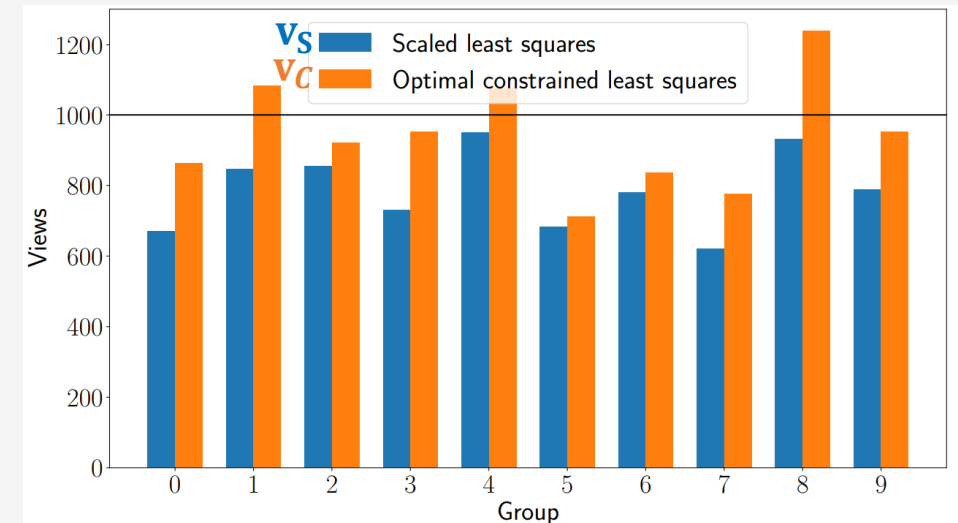
Note that: $0.8 \times \mathbf{1}^T \mathbf{x}^* = 0.8 \times 1605 = 1284$

$\mathbf{x}_s^* = 0.8\mathbf{x}^*$ and $\mathbf{v}_s = 0.8A\mathbf{x}^*$

- **Constrained Least squares**

$$\begin{aligned} \min & \left\| A\mathbf{x} - \mathbf{v}^{des} \right\|^2 \\ \text{s.t. } & \mathbf{1}^T \mathbf{x} = B \end{aligned}$$

- **Optimal spending:** $\mathbf{x}_c^* = (315, 110, 859)^T$
- **Optimal views:** $\mathbf{v}_c = A^T \mathbf{x}_c^*$
- **RMS Error:** 16.10%



Least norm problem *with equality constraints*

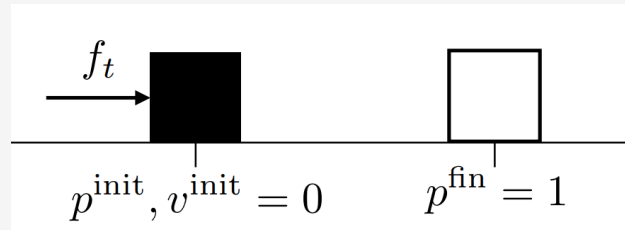
- This is a **special case** of constrained least squares with $A = I$ and $b = 0$

$$\begin{array}{ll} \min ||Ax - b||^2 & \longrightarrow \min ||x||^2 \\ \text{s.t. } Cx = d & \text{s.t. } Cx = d \end{array}$$

- Find the smallest vector that satisfies the set of linear equations**

Example – Force sequence

- Move a unit mass on a frictionless surface.
- **Starting at position** $p^{init} = 0$ and with **initial velocity** $v^{init} = 0$.
- In 10 seconds, the mas should be in **final position** $p^{fin} = 1$ with **final velocity** $v^{fin} = 0$



We can apply a force f_t at the t -th second $\rightarrow$ **force vector** $f = (f_1, f_2, \dots, f_{10})^T$

Newton's law gives us the final velocity and position of the mass:

$$v^{fin} = f_1 + f_2 + \dots + f_{10}$$
$$p^{fin} = (19/2)f_1 + (17/2)f_2 + \dots + (1/2)f_{10}$$

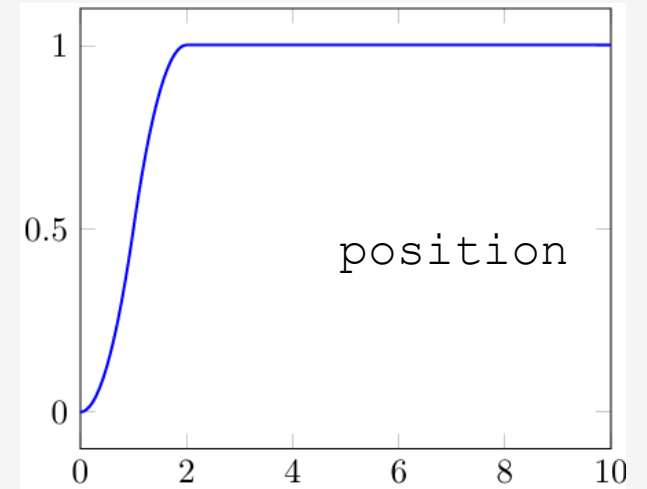
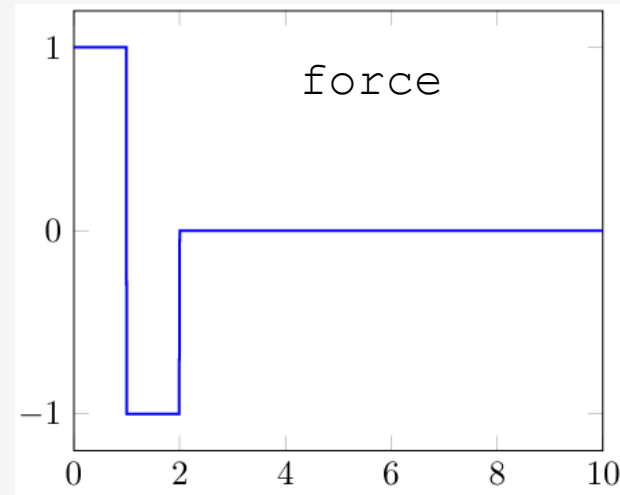
Goal: compute the forces $f = (f_1, f_2, \dots, f_{10})^T$ such that $v^{fin} = 0$ and $p^{fin} = 1$

Example – Force sequence

Solution #1: Bang-Bang solution

$$f^{bb} = (1, -1, 0, \dots, 0)^T$$

$$\|f^{bb}\|^2 = 2$$

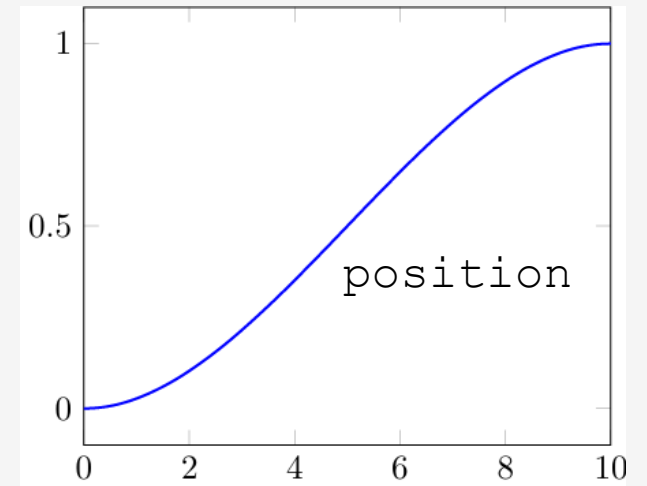
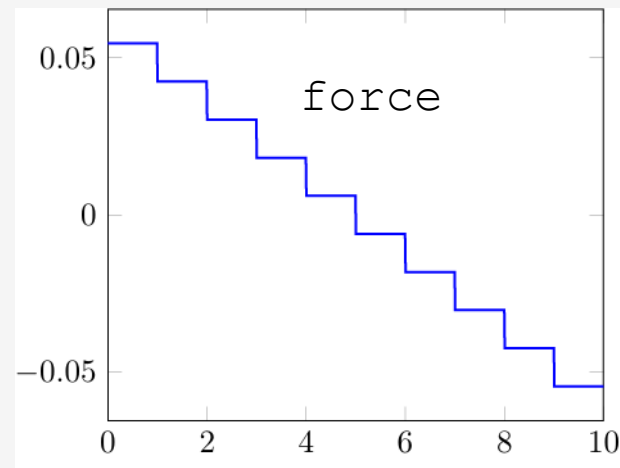


Solution #2: Least norm solution

$$\min \|f\|^2 = f_1^2 + f_2^2 + \dots + f_{10}^2$$

$$\text{s.t. } \begin{bmatrix} 1 & 1 & \dots & 1 \\ (19/2) & (17/2) & \dots & (1/2) \end{bmatrix} f = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\|f^{ln}\|^2 = 0.012 \rightarrow \text{much cheaper effort}$$



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- **Solving constrained least squares problems**

Optimality conditions via Calculus

Constrained least squares problem

$$\min f(x) = \|Ax - b\|^2$$

$$\text{s.t. } Cx = d$$



$$\min f(x) = \|Ax - b\|^2$$

$$\text{s.t. } c_i^T x = d_i, i = 1, \dots, p$$

Lagrangian function

$$L(x, z) = f(x) + z_1(c_1^T x - d_1) + \dots + z_p(c_p^T x - d_p)$$

z_i : **Lagrange multiplier** of constraint i

Vector form:

$$L(x, z) = f(x) + (Cx - d)^T z$$

$z \in R^p$: **Lagrange multiplier vector**

Optimality conditions: $\min L(x, z)$

$$\frac{\partial L}{\partial x_i}(x, z) = 0, \quad i = 1, \dots, n$$

$$\frac{\partial L}{\partial z_k}(x, z) = 0, \quad k = 1, \dots, p$$

**System of $n + p$ equations
in $n + p$ unknowns**

Optimality conditions via Calculus

$$f(x) = \|Ax - b\|^2 = x^T A^T A x - 2(A^T b)^T x + b^T b$$

$$L(x, z) = x^T A^T A x - 2(A^T b)^T x + b^T b + z_1(c_1^T x - d_1) + \cdots + z_p(c_p^T x - d_p)$$

Optimality conditions

Expanded form

For, $i = 1, \dots, n$:

$$\frac{\partial L}{\partial \mathbf{x}_i}(x, z) = 2 \sum_{j=1}^n (A^T A)_{ij} x_j - 2(A^T b)_i + \sum_{k=1}^p (c_k)_i z_k = 0 \quad \longrightarrow \quad \overbrace{2(A^T A)x - 2A^T b + C^T z = 0}$$

For, $k = 1, \dots, p$:

$$\frac{\partial L}{\partial \mathbf{z}_k}(x, z) = c_k^T x - d_k = 0, \quad \longrightarrow \quad \overbrace{Cx - d = 0}$$

Vector form

$$\nabla_{\mathbf{x}} L(x, z) = 0$$

$$\nabla_{\mathbf{z}} L(x, z) = 0$$

Optimality conditions via Calculus

Optimality conditions (vector form)

$$\nabla_{\mathbf{x}} L(\mathbf{x}, \mathbf{z}) = \mathbf{0} \Leftrightarrow 2(A^T A)\mathbf{x} + C^T \mathbf{z} = 2A^T \mathbf{b}$$

$$\nabla_{\mathbf{z}} L(\mathbf{x}, \mathbf{z}) = \mathbf{0} \Leftrightarrow C\mathbf{x} = \mathbf{d}$$

The Optimality conditions are also called the **Karush-Kuhn-Tucker (KKT) conditions**

The **KKT conditions** are often described in the **matrix-vector form**:

$$\begin{bmatrix} 2A^T A & C^T \\ C & 0 \end{bmatrix} \begin{pmatrix} \mathbf{x} \\ \mathbf{z} \end{pmatrix} = \begin{pmatrix} 2A^T \mathbf{b} \\ \mathbf{d} \end{pmatrix} \quad (n+p) \times (n+p) \text{ square system of linear equations}$$

Remark: The KKT conditions are an **extension of the normal equations to constrained problems**

The **normal equations** are: $A^T A \mathbf{x} = A^T \mathbf{b}$ and are the optimality conditions of the **unconstrained least squares**

Invertibility of the KKT matrix

In general, the KKT matrix **may not be positive definite** and therefore **may not be invertible**

$$\begin{bmatrix} 2A^T A & C^T \\ C & 0 \end{bmatrix} \begin{pmatrix} x \\ z \end{pmatrix} = \begin{pmatrix} 2A^T b \\ d \end{pmatrix}$$

Theorem: the KKT matrix **is invertible** if and only if the following conditions are satisfied:

1. The matrix C has **linear independent rows** (i.e., $p \leq n$ and C is a wide matrix)
2. The matrix $\begin{bmatrix} A \\ C \end{bmatrix}$ has **linear independent columns** (i.e., $m + p \geq n$, $\begin{bmatrix} A \\ C \end{bmatrix}$ is a tall matrix)

Proof: see page 345 at VLMS book!

Complexity of the KKT system:

- **Factor+Solve:** $2mn^2 + (2/3)(n + p)^3 + 2(n + p)^2 \approx 2mn^2$ flops
- **For a new b (pre-factored):** $2mn + 2(n + p)^2 \approx 2mn$ flops

**Same as in the
unconstrained case**

Optimal solution from the KKT solutions

Let (x^*, z^*) be the solution of the KKT system $\begin{bmatrix} 2A^T A & C^T \\ C & 0 \end{bmatrix} \begin{pmatrix} x^* \\ z^* \end{pmatrix} = \begin{pmatrix} 2A^T b \\ d \end{pmatrix} \Leftrightarrow \begin{matrix} 2A^T A x^* + C^T z^* = 2A^T b \\ C x^* = d \end{matrix}$

CLAIM: x^* is the optimal solution of the constrained least squares problem

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We take any $x \in \mathcal{F}$ (i.e., $Cx = d$) and any $z \in R^p$ and will show: $\|Ax^* - b\|^2 \leq \|Ax - b\|^2$

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$$\|Ax - b\|^2 = \|Ax - Ax^* - (b - Ax^*)\|^2 = \|A(x - x^*)\|^2 + \|b - Ax^*\|^2 - 2(A(x - x^*))^T (b - Ax^*)$$

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Expanding the last term: $2(A(x - x^*))^T (b - Ax^*) = \dots = 2(x - x^*)^T (A^T b - A^T Ax^*)$

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But both x and x^* are **feasible points**, so $Cx = Cx^* = d \Rightarrow 2(A(x - x^*))^T (b - Ax^*) = 0$

Optimal solution from the KKT solutions

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But both x and x^* are **feasible points**, so $Cx = Cx^* = d \Rightarrow 2(A(x - x^*))^T (b - Ax^*) = 0$

Hence: $\|Ax - b\|^2 = \|A(x - x^*)\|^2 + \|b - Ax^*\|^2 \geq \|b - Ax^*\|^2 \Rightarrow \|b - Ax^*\|^2 \leq \|Ax - b\|^2, \forall x \in \mathcal{F}$
 ≥ 0 x^* is optimal

- **Portfolio optimization**

(Harry Markowitz, 1953 paper → Nobel Laureate in Economics, 1990)

Portfolio allocation weights

We have a **capital** of V dollars and want to **invest** it in n different assets over periods $t = 1, \dots, T$.

$w \in R^n$: the vector of the fractions allocated in each of the n assets

Individual investment in asset i (in dollars): Vw_i

Normalization: $\mathbf{1}^T w = 1$

If $w_j < 0$ then you have a **short position** (you borrowed money, and must return it some time)

Example: $w = (-0.2, 0.0, 1.2)^T$

- Short position on Asset 1 (you borrow shares of Asset 1 worth $0.2V$ and sell them immediately)
- No investment on Asset 2
- Long position on Asset 3 (since you borrowed $0.2V$ you can now invest more than you initial capital in asset 3)

Leverage, long-only portfolios, and cash

Leverage (or Gross Exposure):

$$L = |w_1| + |w_2| + \cdots |w_n| = \|w\|_1$$

If $L = 1$, then all weights are nonnegative ("long only portfolio")

If $L > 1$, then there is at least one negative weight ("long-short portfolio")

Uniform portfolio:

$$w = \mathbf{1}/n = \left(\frac{1}{n}, \frac{1}{n}, \dots, \frac{1}{n}\right)^T$$

Risk-free asset

We often assume the n -th asset is risk-free (e.g., cash, CDs)

If $w = e_n = (0, 0, \dots, 0, 1)^T \rightarrow$ all our money in cash, CDs.

Return matrix

Assume that you hold a constant portfolio with n assets and with weights w over T periods.

The return matrix is a $T \times n$ matrix of the form

$$\mathcal{R} = \begin{matrix} & \begin{matrix} AAPL & GOOG & MMM & US\$ \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ \vdots \\ T \end{matrix} & \begin{bmatrix} 0.00219 & 0.0006 & -0.00113 & 0.0005 \\ 0.00744 & -0.00894 & -0.00019 & 0.0005 \\ 0.01488 & -0.00215 & 0.00433 & 0.0005 \\ \vdots & \vdots & \vdots & \vdots \\ 0.0085 & 0.0102 & -0.0067 & 0.0005 \end{bmatrix} \end{matrix} \in \mathbb{R}^{T \times n}, \quad \mathcal{R}_{tj} = \frac{p_{tj}^{final} - p_{tj}^{initial}}{p_{tj}^{initial}}$$

$\tilde{r}_3$ points to the element 0.01488 in the matrix.

Column interpretation: column j represents the time series returns of asset j

Row interpretation: row t is the returns of the assets over period t (that is, $\tilde{r}_t$)

Portfolio returns (over T time periods): $r = \mathcal{R}w$

NOTE: the last column represents an investment in the **risk-free asset** and therefore is $\mathbf{1}\mu^{rf}$, where μ^{rf} is the interest rate of the risk-free asset

Return over a time period t

Asset returns: $\tilde{r}_t \in R^n \rightarrow$ The (fractional) return vector of the return of each asset over period t
(the t -th row of the return matrix $\mathcal{R}$)

Example:

$\tilde{r}_t = (0.01, -0.023, 0.02)^T$ as fractional returns --- $\tilde{r}_t = (1\%, -2.3\%, 2\%)^T$ as percentage returns

Portfolio return over period t : $r_t = \tilde{r}_t^T w$

Value of investment on asset j at period $t + 1$: $V_{t+1}^{(j)} = V_t w_j + V_t w_j \mathcal{R}_{tj} = V_t w_j (1 + \mathcal{R}_{tj})$

Total portfolio value at period $t + 1$:

$$V_{t+1} = \sum_{j=1}^n V_{t+1}^{(j)} = \sum_{j=1}^n (V_t w_j + V_t w_j \mathcal{R}_{tj}) = \dots = V_t (1 + \tilde{r}_t^T w)$$

Returns over multiple periods

$r \in R^T$ is the time series of our **portfolio returns** over T periods (all time periods).
Number of time periods; NOT Transpose

Average portfolio return (or just return)

$$\text{avg}(r) = \frac{1}{T} (r_1 + \dots + r_T) = \frac{1}{T} \mathbf{1}^T r$$

Portfolio risk (i.e., standard deviation)

$$\text{std}(r) = \|r - \text{avg}(r)\|_2 / \sqrt{T}$$

Total portfolio value (after the end of period T):

$$V_{T+1} = V_T(1 + r_T) = V_{T-1}(1 + r_{T-1})(1 + r_T) = \dots = V_1(1 + r_1) \dots (1 + r_T)$$

$$\approx V_1 + V_1(r_1 + r_2 + \dots + r_T) = V_1 + V_1 T \frac{1}{T} (r_1 + \dots + r_T) = V_1 + V_1 T \text{avg}(r)$$

(when all returns r_i are small)

Observation: to **maximize** our **total final portfolio value** we must have **large** $\text{avg}(r)$

Annualized return and risk

Mean return ($\text{avg}(r)$) and **risk** ($\text{std}(r)$) are expressed in **annualized form**.

Assume there are P **trading periods per year**.

(For instance, we can trade stocks $P = 250$ days per year)

Annualized return:

$$P \text{avg}(r)$$

Annualized risk:

$$\sqrt{P} \text{std}(r)$$

Why annualize? Because it allows for "apples-to-apples" comparisons of performance across investments with different holding periods (e.g., daily, monthly, or multi-year) and diverse asset classes.

Portfolio optimization

Main aim: how should we choose the portfolio weights
(aka, capital allocations, investment fractions)

It depends on your investment objectives

$$r = \mathcal{R}w$$

For high mean return

Maximize $avg(r)$

For low risk

Minimize $std(r)$

Data

- We use **historical prices** to compute realized/past asset returns (future returns are unknown!)
- We compute **optimal weights** (w^*) based on the past returns
- **True goal:** we **hope** those weights will also work well for **future** returns (just like in data fitting)

Portfolio optimization

Goal: minimize risk given target level of return ρ

$$\begin{aligned} \min \quad & \text{std}(\mathcal{R}_w)^2 = \frac{1}{T} \|\mathcal{R}_w - \rho \mathbf{1}\|_2^2 \\ \text{s.t.} \quad & \text{avg}(\mathcal{R}_w) = \rho \\ & \mathbf{1}^T w = 1 \end{aligned}$$

For different target returns ρ , the optimal solutions w are **Pareto optimal**

Our question:

What would have been the optimal constant allocation (investment) w ,
if we had known the future returns?

Some possible allocations (investments)

Annual return: 1% (everything on the risk-free asset)

$$w = e_n = (0.0, 0.0, \dots, 0.0, 1.0)^T$$

Annual return: 13%

$$w = (0.02, -0.07, \dots, 0.56)^T$$

Annual return: 25%

$$w = (0.05, -0.143, \dots, 0.12)^T$$

Higher returns demand:

- Larger allocations on assets with **higher risk**
- **Larger short positions** (borrow money to invest - leverage)

Portfolio optimization

Portfolio optimization problem $\longrightarrow$

$$\begin{aligned} \min \quad & \text{std}(\mathcal{R}w)^2 = \frac{1}{T} \|\mathcal{R}w - \rho \mathbf{1}\|^2 \\ \text{s.t.} \quad & \text{avg}(\mathcal{R}w) = \rho \Rightarrow \mu^T w = \rho \\ & \mathbf{1}^T w = 1 \end{aligned}$$

Constrained least squares problem

$$\begin{aligned} \min \quad & \|\mathcal{R}w - \rho \mathbf{1}\|^2 \\ \text{s.t.} \quad & \begin{bmatrix} \mu^T \\ \mathbf{1}^T \end{bmatrix} w = \begin{bmatrix} \rho \\ 1 \end{bmatrix} \Rightarrow \begin{aligned} \mu^T w &= \rho \\ \mathbf{1}^T w &= 1 \end{aligned} \end{aligned}$$

Lagrangian function:

$$L(x, z_1, z_2) = \|\mathcal{R}w - \rho \mathbf{1}\|^2 + z_1(\mu^T w - \rho) + z_2(\mathbf{1}^T w - 1)$$

KKT conditions:

$$\begin{bmatrix} 2\mathcal{R}^T \mathcal{R} & \mu & 1 \\ \mu^T & 0 & 0 \\ \mathbf{1}^T & 0 & 0 \end{bmatrix} \begin{bmatrix} w \\ z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} 2T\mu\rho \\ \rho \\ 1 \end{bmatrix}$$

(do the algebra!)

Optimal portfolios

The optimal portfolio is an affine function of the target returns ρ

Optimal portfolio for a given target return ρ :

$$\begin{bmatrix} w \\ z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} 2\mathcal{R}^T \mathcal{R} & \mu & 1 \\ \mu^T & 0 & 0 \\ 1^T & 0 & 0 \end{bmatrix}^{-1} \begin{bmatrix} 2T\mu\rho \\ \rho \\ 1 \end{bmatrix}$$

The two fund theorem:

$$\begin{bmatrix} w \\ z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} 2\mathcal{R}^T \mathcal{R} & \mu & 1 \\ \mu^T & 0 & 0 \\ 1^T & 0 & 0 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + \rho \begin{bmatrix} 2\mathcal{R}^T \mathcal{R} & \mu & 1 \\ \mu^T & 0 & 0 \\ 1^T & 0 & 0 \end{bmatrix}^{-1} \begin{bmatrix} 2T\mu \\ 1 \\ 0 \end{bmatrix} = w_0 + \rho v$$

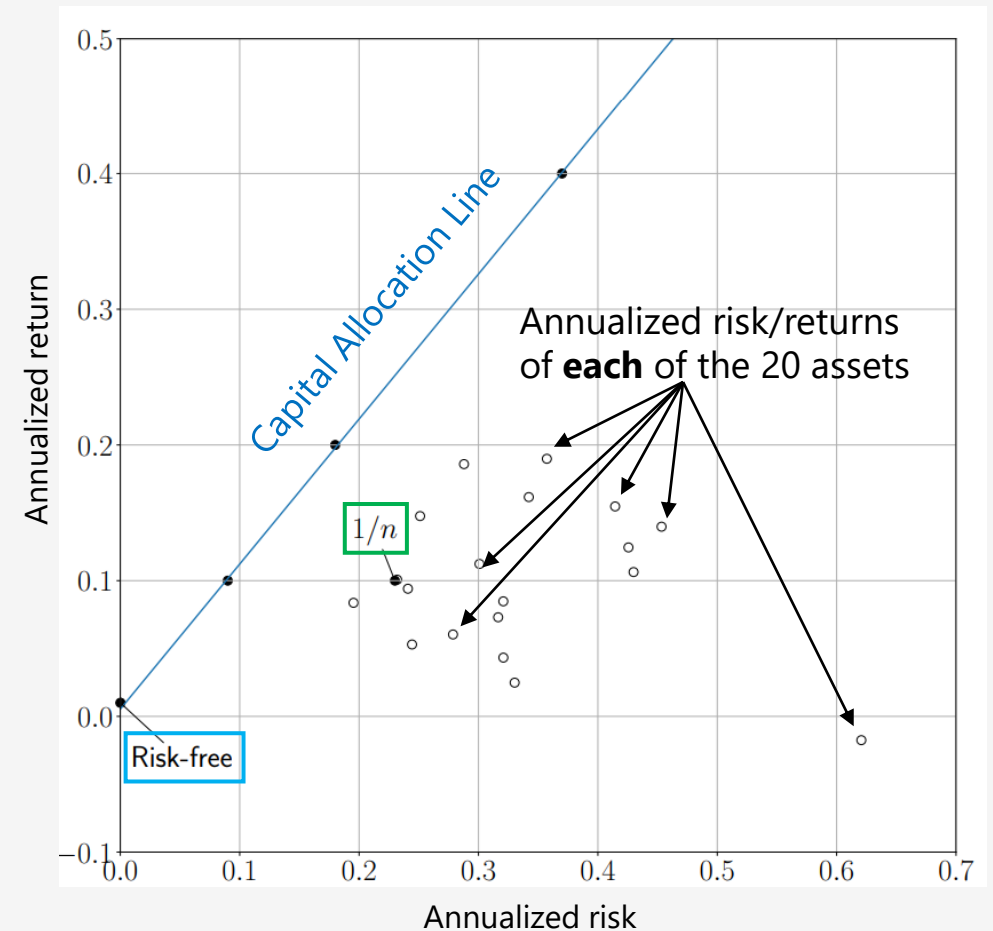
affine function of ρ

where w_0 is the **risk-free allocation** (when $\rho = 0$) and v is the allocation on the **risky assets**

Numerical Example

We can invest in **20 assets**, and we have **2000 days historical data**

- **Optimal portfolios are on the straight line**
- Line start at the **risk-free-only portfolio** ($\rho = 0$)
- The **1/n portfolio** performs much better than the single portfolios



The big assumption (and deception)

By using historical returns, we have assumed that the future returns will look like past ones.

Common practice:

- Compute w by using historical data of the past 2-5 years.
- Then use w for the next 6 months
- After 6 months compute a new w using more recent historical data

But in general, the assumption is not true in many cases.

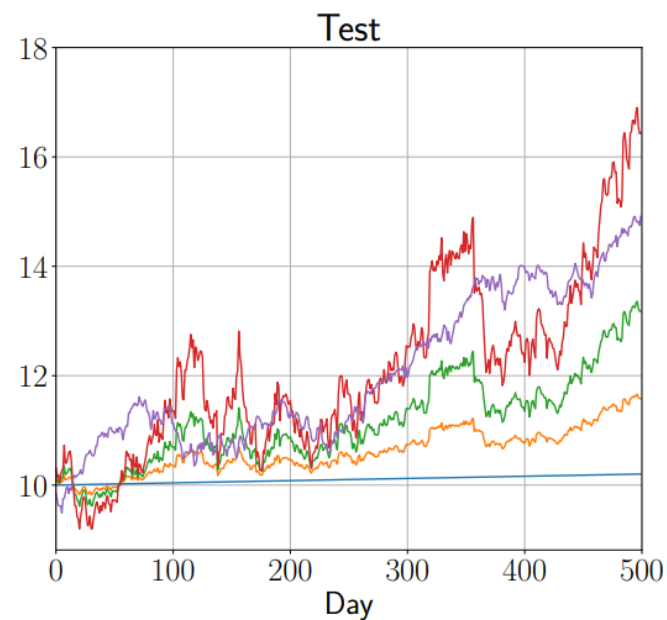
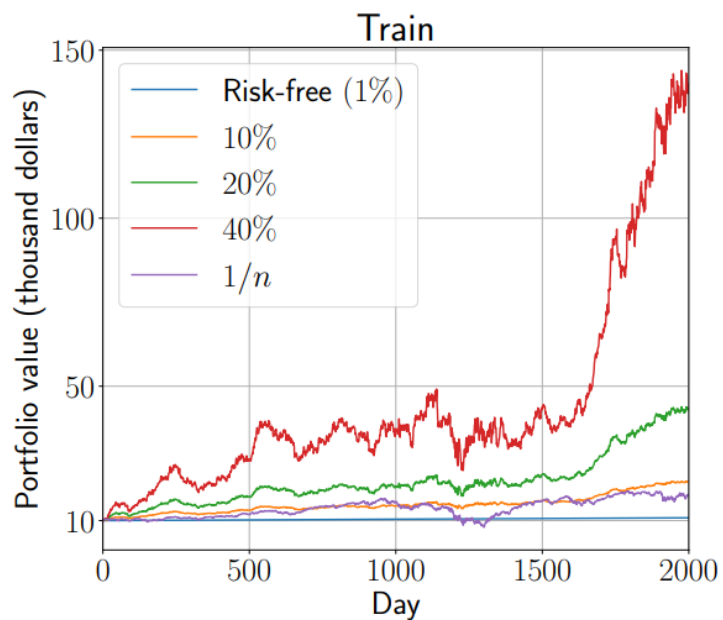
- Especially when global events (e.g., COVID) happen that are not captured in historical data

The **SEC** requires the disclaimer to be included in investment products:

“Past performance is not guarantee for future results!”

Total portfolio value

	Return		Risk		Leverage
	Train	Test	Train	Test	
Risk-free (1%)	0.01	0.01	0.00	0.00	1.00
10%	0.10	0.08	0.09	0.07	1.96
20%	0.20	0.15	0.18	0.15	3.03
40%	0.40	0.30	0.37	0.31	5.48
1/n	0.10	0.21	0.23	0.13	1.00



Build your own quantitative hedge fund

Rolling portfolio optimization

For each time-period t , compute weights w_t by looking N periods back

$$r_{t-1}, r_{t-2}, \dots, r_{t-N}$$

Other variations

1. Update w every K periods (e.g., monthly, quarterly, ...)
2. Add a secondary objective $\lambda ||w_t - w_{t-1}||^2$ to discourage turnover and reduce transaction costs
3. Add logic to detect when the future is likely to not look like the past
4. Add "signals" that predict future returns of assets (Twitter sentiment analysis)

References

- S. Boyd, L. Vandenberghe: Introduction to Applied Linear Algebra – Vectors, Matrices, and Least Squares:
 - Ch. 16
 - Ch. 17 → Portfolio optimization

Next lecture

- Linear optimization
